Nonlocal one-loop form factors of the spectral action with Standard Model content

David Alfyorov[1]

[1] *Independent researcher*[*]

We compute the matter-induced nonlocal one-loop curvature-squared form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2, \xi)$ relevant to the spectral action $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$, for Standard Model field content: 4 real scalars (Higgs doublet), 45/2 Dirac-equivalent fermions (3 generations), and 12 gauge bosons ($\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$). Using Barvinsky–Vilkovisky covariant perturbation theory [1, 2] and the Codello–Zanusso nonlocal heat kernel [3], we derive closed-form reduced kernels $h_C^{(s)}(x)$ and $h_R^{(s)}(x)$ for each spin sector ($s = 0, 1/2, 1$) in the Weyl basis $\{C^2, R^2\}$, with step-by-step assembly from the five universal CZ structure functions. For spin-1, we provide the complete Faddeev–Popov ghost derivation with two minimally coupled scalar ghosts. The Standard Model assembly yields $\alpha_C = 13/120$ (Weyl-squared, ξ-independent) and $\alpha_R(\xi) = 2(\xi - 1/6)^2$ (R^2, Higgs non-minimal coupling only), reproducing all known local limits [4–7]. After removing apparent singularities at $z = 0$, the reduced form factors are proven to be entire functions of $z = \Box/\Lambda^2$ of order 1 and type 1/4; consequently, the form factors themselves introduce no propagator poles. Ghost poles arise from zeros of the dressed propagator, which we derive from the full linearized quadratic action using Barnes–Rivers spin projectors [8, 9]. This yields effective masses $m_2 = \Lambda\sqrt{60/13}$ (spin-2) and $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$ (spin-0), with the scalar mode decoupling at conformal coupling $\xi = 1/6$. All results are independently verified by triple computer algebra (SymPy, cadabra2 [10], FORM [11]), arbitrary-precision numerical evaluation (≥ 100 digits), and Lean 4 formal proofs for the coefficient arithmetic.

I. INTRODUCTION

The spectral action principle [12–14] provides a geometric origin for both gravitational and gauge interactions: the bosonic action is given by $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$, where D is the Dirac operator of a noncommutative geometry, Λ is a cutoff scale, and f is a positive even function. At the classical level, the heat kernel expansion of this trace reproduces the Einstein–Hilbert action, cosmological constant, and Yang–Mills terms from the first few Seeley–DeWitt coefficients [15, 16].

Beyond the leading Seeley–DeWitt approximation, the spectral action generates *nonlocal* curvature-squared terms characterized by momentum-dependent form factors $F_1(\Box/\Lambda^2)$ and $F_2(\Box/\Lambda^2, \xi)$. These form factors encode the *matter-induced* contribution to the one-loop curvature-squared effective action at $\mathcal{O}(\mathcal{R}^2)$: they arise from functional traces over matter fields (scalars, fermions, gauge bosons) propagating on a fixed curved gravitational background. The graviton self-energy contributions, which would constitute the full quantum gravitational effective action, are not computed here; they require a separate treatment of the metric fluctuation operator, its gauge-fixing, and the gravitational ghost sector [7, 17]. Nevertheless, the matter-induced form factors already determine the modified graviton propagator structure, effective masses, and the UV behavior when matter-loop contributions dominate.

The Barvinsky–Vilkovisky (BV) covariant perturbation theory [1, 2, 18, 19] provides the general framework for computing such form factors for any generalized Laplacian $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ on a vector bundle over a curved manifold. The Codello–Zanusso (CZ) diagrammatic heat kernel technique [3] gives explicit expressions for the five independent structure functions $(f_{\mathrm{Ric}}, f_R, f_{RU}, f_U, f_\Omega)$ in terms of a universal master function $\varphi(x)$, which depends only on the proper-time parameter and is independent of the profile function f at $\mathcal{O}(\mathcal{R}^2)$ (see Appendix D for a proof). The Seeley–DeWitt coefficients and their role in spectral geometry are reviewed in [4, 5, 20–22].

The per-spin form factors for individual Laplace-type operators were obtained by BV [2, 18] and CZ [3]; local limits (β-function coefficients) were tabulated for matter fields by CPR [6] and in the textbook by Parker and Toms [23]. One-loop corrections to the spectral action were studied by van Nuland and van Suijlekom [24], and the spectral-action framework for higher-derivative gravity was analyzed by Mistry, Pinzul, and Rachwał [25]. The present contribution assembles the full Standard Model (SM) per-spin form factors into SM-specific quantities (α_C, $\alpha_R(\xi)$, c_1/c_2) that have not been previously reported, proves the entire-function property of the SM totals, and derives the dressed graviton propagator from the complete linearized quadratic action.

The SM particle content consists of 4 real scalars (Higgs doublet), $N_f = 45$ Weyl fermions (equivalently $N_D = 45/2$ Dirac fermions in 3 generations), and 12 gauge bosons of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$. The counting convention follows CPR [6].

The main results are:

- Closed-form expressions for the reduced form factors $h_C^{(s)}(x)$ and $h_R^{(s)}(x)$ for $s = 0, 1/2, 1$ in the Weyl

[*] davidich.alfyorov@gmail.com

basis $\{C^2, R^2\}$, with complete step-by-step derivation of the BV/CZ assembly for each spin sector (Secs. III–V).

- The SM totals $\alpha_C = 13/120$ (ξ-independent, determined by SM content) and $\alpha_R(\xi) = 2(\xi - 1/6)^2$ (depending only on the Higgs non-minimal coupling ξ), with the local R^2 coefficient vanishing at conformal coupling $\xi = 1/6$ (Sec. VII).
- A proof that the reduced form factors, after removal of apparent singularities at $z = 0$, are entire functions of order 1 and type 1/4 (Theorem VIII.1). As a consequence, the form factors introduce no propagator poles; ghost poles arise solely from zeros of the dressed propagator (Sec. VIII).
- The dressed spin-2 and spin-0 propagators $\Pi_{TT}(z)$, $\Pi_s(z, \xi)$, derived from the full linearized quadratic action via Barnes–Rivers projectors, yielding effective masses $m_2 = \Lambda\sqrt{60/13}$, $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$ and the modified Newtonian potential (Sec. X).
- A systematic comparison with BV, CZ, CPR, Parker–Toms, Stelle, and the Donoghue effective field theory [26, 27] (Sec. XI).

The structure of the paper is as follows. Section II establishes the conventions and reviews the BV/CZ formalism, including the f-independence of the form factors at $\mathcal{O}(\mathcal{R}^2)$. Sections III–V derive the form factors for each spin sector with explicit BV assembly. Section VII assembles the SM totals. Section VIII proves the entire-function property. Section X derives the dressed propagator and effective masses. Section IX discusses UV asymptotics. Section XI compares with existing results, including the Donoghue EFT. Section XII discusses implications for asymptotic safety and nonlocal gravity. We conclude in Sec. XIII.

Throughout this paper we use the Euclidean signature $(+,+,+,+)$, natural units $c = \hbar = 1$, and the generalized Laplacian convention $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ of Barvinsky and Vilkovisky [1]. The CZ endomorphism is $\mathbf{U} = -E$. The Lorentzian continuation to $(-,+,+,+)$ signature is discussed in Sec. XII. Convention tables are collected in Appendix A. A summary of the principal results is given in Table II.

II. SETUP AND CONVENTIONS

A. Generalized Laplacian and spectral action

We consider a closed (compact, boundaryless) Riemannian spin 4-manifold (M, g). For each matter field of spin s, the one-loop effective action is

$$\Gamma^{(s)} = \frac{(-1)^{2s}}{2}\,\mathrm{Tr}\ln\Delta^{(s)}\,, \tag{1}$$

where $\Delta^{(s)}$ is the generalized Laplacian of the corresponding kinetic operator, and $(-1)^{2s} = +1$ for bosons, -1 for fermions. At $\mathcal{O}(\mathcal{R}^2)$, the curvature-squared sector of the effective action takes the form

$$\Gamma\Big|_{\mathcal{R}^2} = \int_M \mathrm{d}^4x\,\sqrt{g}\,\left[F_1(z)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + F_2(z,\xi)\,R^2\right], \tag{2}$$

where $z \equiv \Box/\Lambda^2$ and Λ is the spectral cutoff scale. The form factors F_i are the central objects of this paper.

For a generalized Laplacian $\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$ acting on a vector bundle of rank d_V (so that $\mathrm{tr}\,\mathbf{1} = d_V$), the CZ endomorphism is $\mathbf{U} = -E$ and the bundle curvature is $\Omega_{\mu\nu} = [\nabla_\mu, \nabla_\nu]$ acting on sections of the bundle. We write

$$F_i^{(s)}(z) = \frac{h_i^{(s)}(z)}{16\pi^2}\,, \quad i = C, R\,, \tag{3}$$

where $h_C^{(s)}$ and $h_R^{(s)}$ are the *reduced* form factors. Their local limits $\beta_W^{(s)} \equiv h_C^{(s)}(0)$ and $\beta_R^{(s)} \equiv h_R^{(s)}(0)$ are the one-loop β-function coefficients of the C^2 and R^2 operators. In our convention, the fermionic sign from (1) is not applied as an explicit overall factor; it is encoded in the assembly rules through the operator data (Sec. II E). For the SM spins $s = 0, 1/2, 1$, the resulting $h_C^{(s)}(0)$ is positive for bosons and negative for the Dirac fermion, but this pattern does not extend to all spins (see Sec. VI).

B. From heat-kernel kernels to spectral-action form factors

The spectral action $S = \mathrm{Tr}\,f(D^2/\Lambda^2)$ depends on a profile function f. We now make precise how the form factors depend on f, and under what conditions they are f-independent.

If f admits a Laplace representation $f(u) = \int_0^\infty \mathrm{d}\alpha\,\tilde{f}(\alpha)\,e^{-\alpha u}$, then the spectral action at $\mathcal{O}(\mathcal{R}^2)$ becomes

$$S_f\big|_{\mathcal{R}^2} = \frac{1}{16\pi^2}\int_M \mathrm{d}^4x\,\sqrt{g}\,\left[\mathcal{H}_C(z;f)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \mathcal{H}_R(z,\xi;f)\,R^2\right], \tag{4}$$

where the *profile-smeared* form factors are

$$\mathcal{H}_i(z;f) = \int_0^\infty \mathrm{d}\alpha\,\tilde{f}(\alpha)\,h_i(\alpha z)\,, \quad i = C, R\,. \tag{5}$$

Here h_i denotes the *reduced kernel* corresponding to the exponential profile $f(u) = e^{-u}$ (i.e., $\tilde{f}(\alpha) = \delta(\alpha - 1)$). The reduced kernels, built from the master function $\varphi(x)$, are universal — they depend only on the Laplace-type operator and not on the spectral profile. The profile dependence enters solely through the smearing (5).

Remark II.1. All explicit closed formulas in this paper are reduced kernels h_i. They coincide with the spectral-action form factors only for the exponential profile, or more generally for any delta-supported proper-time measure.

A direct consequence of (5) is the profile dependence of the local coefficients:

$$\mathcal{H}_i(0;f) = h_i(0)\, f(0)\,, \tag{6}$$

since $\int_0^\infty \tilde{f}(\alpha)\,\mathrm{d}\alpha = f(0)$. Hence the *ratios* of local curvature-squared couplings (e.g. c_1/c_2) are profile-independent, while the overall normalization is proportional to $f(0)$. For the SM local coefficients:

$$\alpha_C^{\rm loc} = \frac{13}{120}\, f(0)\,, \qquad \alpha_R^{\rm loc} = 2\, f(0)\left(\xi - \frac{1}{6}\right)^2 . \tag{7}$$

Throughout this paper, we set $f(0) = 1$ (exponential profile) unless otherwise stated.

C. Master function

All form factors are built from the universal master function

$$\varphi(x) = \int_0^1 \mathrm{d}\alpha\, e^{-\alpha(1-\alpha)x}\,, \tag{8}$$

with the closed form

$$\varphi(x) = e^{-x/4}\sqrt{\frac{\pi}{x}}\ \mathrm{erfi}\left(\frac{\sqrt{x}}{2}\right) . \tag{9}$$

Key properties: $\varphi(0) = 1$, $\varphi'(0) = -1/6$, and Taylor coefficients $a_n = (-1)^n\, n!/(2n+1)!$, so that $\varphi(x) = \sum_{n=0}^\infty a_n\, x^n$ with infinite radius of convergence (the ratio test gives $|a_{n+1}/a_n| \to 0$). In particular, φ is an entire function of order 1 and type 1/4 (Fig. 1). The large-x asymptotic expansion is

$$\varphi(x) = \frac{2}{x} + \frac{4}{x^2} + \frac{24}{x^3} + \frac{240}{x^4} + \mathcal{O}(x^{-5})\,, \tag{10}$$

which is an asymptotic (divergent) series, Borel-summable to the exact φ.

D. CZ structure functions

The five CZ structure functions [3] are:

$$f_{\rm Ric}(x) = \frac{1}{6x} + \frac{\varphi - 1}{x^2}\,, \tag{11a}$$

$$f_R(x) = \frac{\varphi}{32} + \frac{\varphi}{8x} - \frac{7}{48x} - \frac{\varphi - 1}{8x^2}\,, \tag{11b}$$

$$f_{RU}(x) = -\frac{\varphi}{4} - \frac{\varphi - 1}{2x}\,, \tag{11c}$$

$$f_U(x) = \frac{\varphi}{2}\,, \tag{11d}$$

$$f_\Omega(x) = -\frac{\varphi - 1}{2x}\,. \tag{11e}$$

Their local limits are $f_{\rm Ric}(0) = 1/60$, $f_R(0) = 1/120$, $f_{RU}(0) = -1/6$, $f_U(0) = 1/2$, $f_\Omega(0) = 1/12$.

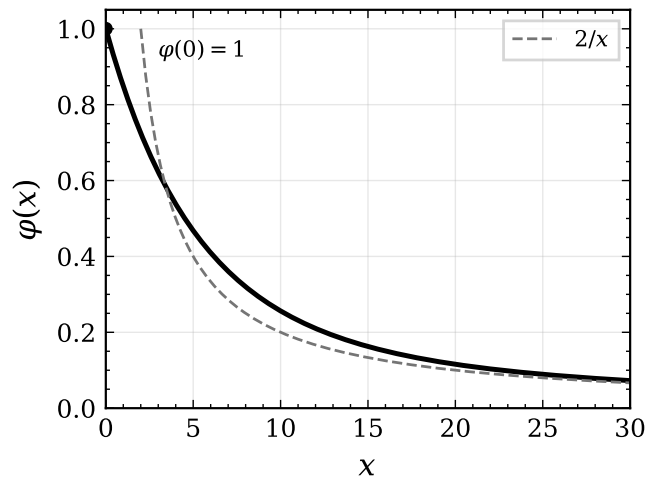

FIG. 1. The master function $\varphi(x)$, where $x = \Box/\Lambda^2$. Starting from $\varphi(0) = 1$, the function decays monotonically and approaches the asymptotic form $2/x$ (dashed) for $x \gg 1$.

E. Weyl basis assembly

The BV/CZ expansion of the $\mathcal{O}(\mathcal{R}^2)$ heat kernel trace for a single field with operator data $(d_V, \mathbf{U}, \Omega_{\mu\nu})$ contains five universal structure functions [2, 3]. After the proper-time integration, the reduced kernels h_C, h_R are assembled from:

$$\begin{aligned}\frac{1}{16\pi^2}\left[h_C\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + h_R\, R^2\right] \longleftarrow\ & d_V\, f_{\rm Ric}\, R_{\mu\nu}R^{\mu\nu} + d_V\, f_R\, R^2 \\ & + R\, f_{RU}\ \mathrm{tr}(\mathbf{U}) + \mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U}) \\ & + \mathrm{tr}(\Omega_{\mu\nu}\, f_\Omega\, \Omega^{\mu\nu})\,,\end{aligned} \tag{12}$$

where the conversion from the right-hand side (CZ basis) to the left-hand side (Weyl basis) uses (13). The statistics sign $(-1)^{2s}$ is *not* applied as an extra overall factor; it is already encoded in the operator data (specifically, in the sign and magnitude of $\mathrm{tr}(\Omega^2)$ and $\mathbf{U}$) for each spin sector. The five CZ functions act on the *spacetime* argument $x = \Box/\Lambda^2$, while the traces are over *bundle* indices only. The assembly into the Weyl basis $\{C^2, R^2\}$ proceeds by converting the curvature invariants using the Gauss–Bonnet identity in $d = 4$ (modulo the topological Euler density $E_4 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2$):

$$R_{\mu\nu}R^{\mu\nu} = \tfrac{1}{2}C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \tfrac{1}{3}R^2\,, \tag{13a}$$

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \tfrac{1}{3}R^2\,. \tag{13b}$$

The key step, which differs between spin sectors, is the evaluation of $\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U})$ and $\mathrm{tr}(\Omega\, f_\Omega\, \Omega)$:

- If $\mathbf{U} \propto R \cdot \mathbf{1}$ (scalar or proportional to the identity, as for spin-0 and spin-1/2), then $\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U}) \propto R^2 \cdot f_U$, contributing only to the R^2 channel.
- If $\mathbf{U}$ is matrix-valued (e.g., $U^\mu{}_\nu = R^\mu{}_\nu$ for spin-1), then $\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U}) = f_U\, R_{\mu\nu}R^{\mu\nu}$, which contributes to *both* C^2 and R^2 via (13a).

This distinction is the source of the nontrivial assembly structure for higher spins and will be exhibited explicitly in Secs. III–V.

III. SPIN-0: REAL SCALAR

A. Operator data

For a real scalar field with non-minimal coupling $\xi R\phi^2/2$, the generalized Laplacian is $\Delta^{(0)} = -\Box + \xi R$ (i.e., $E = -\xi R$ in the BV convention $\Delta = -(\Box + E)$), giving operator data:

$$d_V = 1\,, \quad \mathbf{U} = \xi R\,, \quad \Omega_{\mu\nu} = 0\,. \tag{14}$$

The endomorphism $\mathbf{U} = \xi R$ is scalar-valued (proportional to the identity on the trivial bundle), and the bundle curvature vanishes since the scalar field carries no internal indices.

B. BV/CZ assembly

The five terms in (12) give:

Term	Curvature structure	Invariant
$d_V\, f_{\rm Ric}\, R_{\mu\nu}R^{\mu\nu}$	$1 \cdot f_{\rm Ric} \cdot R_{\mu\nu}R^{\mu\nu}$	$R_{\mu\nu}R^{\mu\nu}$
$d_V\, f_R\, R^2$	$1 \cdot f_R \cdot R^2$	R^2
$R\, f_{RU}\ \mathrm{tr}(\mathbf{U})$	$\xi\, f_{RU}\, R^2$	R^2
$\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U})$	$\xi^2\, f_U\, R^2$	R^2
$\mathrm{tr}(\Omega\, f_\Omega\, \Omega)$	0	—

Since $\mathbf{U} = \xi R$ is scalar-valued, $\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U}) = \xi^2 R^2\, f_U$ contributes only to R^2, not to C^2.

Converting $R_{\mu\nu}R^{\mu\nu}$ to the Weyl basis via (13a):

$$h_C^{(0)}(x) = \tfrac{1}{2}\, f_{\rm Ric}(x) = \frac{1}{12x} + \frac{\varphi - 1}{2x^2}\,, \tag{15a}$$

$$h_R^{(0)}(x,\xi) = \tfrac{1}{3} f_{\rm Ric}(x) + f_R(x) + \xi\, f_{RU}(x) + \xi^2\, f_U(x)\,. \tag{15b}$$

Lemma III.1 (ξ-independence of $h_C^{(0)}$)**.** *The Weyl coefficient $h_C^{(0)}$ is independent of ξ. This holds because $\mathbf{U} = \xi R$ is proportional to the scalar curvature, which is the trace of the Ricci tensor: $R = g^{\mu\nu}R_{\mu\nu}$. The Weyl tensor is traceless ($g^{\mu\nu}C_{\mu\alpha\nu\beta} = 0$), so only $R_{\mu\nu}R^{\mu\nu}$ appears in the C^2 channel, and $R_{\mu\nu}R^{\mu\nu}$ comes solely from Term 1 (the f_{Ric} term), which is ξ-independent.*

C. Pole removability

Lemma III.2 (Removability of $h_C^{(0)}$ at $x = 0$)**.** *The apparent pole of $h_C^{(0)}(x) = 1/(12x) + (\varphi - 1)/(2x^2)$ at $x = 0$ is removable.* Proof. *Using the shifted-index formula $h_C^{(0)}(x) = \sum_{n\geq 0} \frac{1}{2} a_{n+2}\, x^n$ with $a_n = (-1)^n n!/(2n+1)!$, the poles are manifestly absent. The leading term is $\frac{1}{2}a_2 = 1/120$, confirming analyticity at $x = 0$. The Taylor expansion is $h_C^{(0)}(x) = 1/120 - x/1680 + \mathcal{O}(x^2)$.*

D. Local limits

$$\beta_W^{(0)} = h_C^{(0)}(0) = \frac{1}{120}\,, \qquad \beta_R^{(0)}(\xi) = h_R^{(0)}(0,\xi) = \frac{1}{2}\left(\xi - \frac{1}{6}\right)^2\,. \tag{16}$$

The R^2 coefficient vanishes at conformal coupling $\xi = 1/6$, reflecting the conformal invariance of the massless conformally-coupled scalar in $d = 4$ [23]. For $\xi \neq 1/6$, the nonlocal form factor $h_R^{(0)}(x,\xi)$ is nonzero for all $x > 0$. The individual form factors are plotted in Figs. 2 and 3.

IV. SPIN-1/2: DIRAC FERMION

A. Operator data

For a massless 4-component Dirac fermion, the Lichnerowicz formula gives the squared Dirac operator:

$$\not{D}^2 = -\big(\nabla^\mu\nabla_\mu - \tfrac{R}{4}\big)\,, \tag{17}$$

which is a generalized Laplacian with $E = -R/4$ (note the sign: $\Delta = -(\Box + E)$ in the BV convention), so that the CZ endomorphism is

$$\mathbf{U} = -E = \frac{R}{4}\,\mathbf{1}_4\,. \tag{18}$$

The spinor connection induces the bundle curvature

$$(\Omega_{\mu\nu})^a{}_b = \frac{1}{4}\, R_{\mu\nu\rho\sigma}\, (\gamma^\rho\gamma^\sigma)^a{}_b\,. \tag{19}$$

The relevant traces are (using the $d = 4$ gamma-matrix identities $\mathrm{tr}(\gamma^\mu\gamma^\nu) = 4\, g^{\mu\nu}$ and $\mathrm{tr}(\gamma^{\mu\nu}\gamma^{\rho\sigma}) = 4(\delta^\mu_\rho\delta^\nu_\sigma - \delta^\mu_\sigma\delta^\nu_\rho)$):

$$\mathrm{tr}(\mathbf{1}) = 4\,, \tag{20a}$$

$$\mathrm{tr}(\mathbf{U}) = R\,, \quad \mathrm{tr}(\mathbf{U}^2) = \frac{R^2}{4}\,, \tag{20b}$$

$$\begin{aligned}\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu}) &= \frac{1}{16}\, R_{\mu\nu\rho\sigma}\, R^{\mu\nu\alpha\beta}\ \mathrm{tr}(\gamma^{\rho\sigma}\gamma_{\alpha\beta}) \\ &= -\frac{1}{2}\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\,.\end{aligned} \tag{20c}$$

The sign in (20c) follows from the antisymmetry of $R_{\mu\nu\rho\sigma}$ in the last pair and the trace identity $\mathrm{tr}(\gamma^{\rho\sigma}\gamma_{\alpha\beta}) = 4(\delta^\rho_\alpha\delta^\sigma_\beta - \delta^\rho_\beta\delta^\sigma_\alpha)$; contracting gives $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu\rho\sigma}R^{\mu\nu\sigma\rho} = 2\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, then the factor $1/16 \times 4 \times 2 = 1/2$ with the relative minus from the antisymmetry.

B. BV/CZ assembly

Since $\mathbf{U} = (R/4)\mathbf{1}_4$ is proportional to the identity, $\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U}) = (R^2/4)\, f_U$ contributes only to R^2. The five terms in (12) give:

Term	Contribution (unsigned)	Invariant
Term 1	$4\, f_{\mathrm{Ric}}\, R_{\mu\nu}R^{\mu\nu}$	$R_{\mu\nu}R^{\mu\nu}$
Term 2	$4\, f_R\, R^2$	R^2
Term 3	$f_{RU}\, R^2$	R^2
Term 4	$\frac{1}{4}\, f_U\, R^2$	R^2
Term 5	$-\frac{1}{2}\, f_\Omega\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$

Converting to the Weyl basis via (13):

$$h_C^{(1/2)}(x) = 2\, f_{\mathrm{Ric}} - f_\Omega \tag{21a}$$

$$= \frac{3\varphi - 1}{6x} + \frac{2(\varphi - 1)}{x^2}\,, \tag{21b}$$

$$h_R^{(1/2)}(x) = \tfrac{4}{3} f_{\mathrm{Ric}} + 4 f_R + f_{RU} + \tfrac{1}{4} f_U - \tfrac{1}{6} f_\Omega \tag{21c}$$

$$= \frac{3\varphi + 2}{36x} + \frac{5(\varphi - 1)}{6x^2}\,. \tag{21d}$$

No additional fermionic sign is needed: the negative value $h_C^{(1/2)}(0) = 2f_{\mathrm{Ric}}(0) - f_\Omega(0) = 2/60 - 1/12 = -1/20$ arises directly from the spinor curvature contribution $f_\Omega(0) = 1/12$ outweighing $2f_{\mathrm{Ric}}(0) = 1/30$ in the Weyl projection.

C. Pole removability and local limits

Lemma IV.1 (Removability of $h_C^{(1/2)}$ at $x = 0$)**.** *Using the shifted-index formula* $h_C^{(1/2)}(x) = \sum_{n\geq 0}(\frac{1}{2}a_{n+1} + 2a_{n+2})\, x^n$ *with* $a_n = (-1)^n n!/(2n+1)!$*, the apparent poles are manifestly absent. The leading terms are* $\frac{1}{2}a_1 + 2a_2 = -1/12 + 1/30 = -1/20$ *and* $\frac{1}{2}a_2 + 2a_3 = 1/120 - 1/420 = 1/168$*, giving* $h_C^{(1/2)}(x) = -1/20 + x/168 + \mathcal{O}(x^2)$*.*

The local limits are

$$\beta_W^{(1/2)} = -\frac{1}{20}\,, \qquad \beta_R^{(1/2)} = 0\,, \tag{22}$$

consistent with $2f_{\mathrm{Ric}}(0) - f_\Omega(0) = 2/60 - 1/12 = -1/20$ (negative directly from the curvature combination, no extra fermion sign). The vanishing $\beta_R^{(1/2)} = 0$ reflects the conformal invariance of the massless Dirac action in $d = 4$ [6, 23].

V. SPIN-1: GAUGE BOSON

For a massless gauge boson in the spectral action framework, the generalized Laplacian acts on the tangent bundle (vector representation) with $\mathrm{tr}\,\mathbf{1} = 4$, $\mathbf{U} = R_{\mu\nu}$ (Ricci tensor as endomorphism), and $\Omega_{\mu\nu} = R_{\mu\nu\rho\sigma}$ (Riemann tensor as bundle curvature) [5, 23]. This is the background-field operator for a gauge boson in the de Donder gauge; the terminology "Proca field" sometimes used in the literature is misleading, as the field is massless and retains gauge symmetry.

A crucial difference from the scalar and Dirac cases is that $\mathbf{U}$ is now *matrix-valued*: $U^\mu{}_\nu = R^\mu{}_\nu$, so that $\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U}) = f_U \cdot R_{\mu\nu}R^{\mu\nu}$, which contributes to both C^2 and R^2 channels after the Weyl basis conversion. The five traced CZ terms for the unconstrained vector are:

CZ term	Curvature invariant	Coefficient
$\mathrm{tr}(\mathbf{1})\, f_{\mathrm{Ric}}\, R_{\mu\nu}R^{\mu\nu}$	$4\, f_{\mathrm{Ric}}\, R_{\mu\nu}R^{\mu\nu}$	Term 1
$\mathrm{tr}(\mathbf{1})\, f_R\, R^2$	$4\, f_R\, R^2$	Term 2
$R\, f_{RU}\, \mathrm{tr}(\mathbf{U})$	$f_{RU}\, R^2$	Term 3
$\mathrm{tr}(\mathbf{U}\, f_U\, \mathbf{U})$	$f_U\, R_{\mu\nu}R^{\mu\nu}$	Term 4
$\mathrm{tr}(\Omega\, f_\Omega\, \Omega)$	$-f_\Omega\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	Term 5

where $\mathrm{tr}(\mathbf{U}) = R$ and $\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu}) = -R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ for the tangent bundle. Converting to the $\{C^2, R^2\}$ Weyl basis using $R_{\mu\nu}R^{\mu\nu} = \frac{1}{2}C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{1}{3}R^2$ and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{1}{3}R^2$ (modulo the Euler density), the unconstrained form factors are:

$$h_C^{(\mathrm{unc})}(x) = 2\, f_{\mathrm{Ric}}(x) + \tfrac{1}{2}\, f_U(x) - 2\, f_\Omega(x)\,, \tag{23a}$$

$$h_R^{(\mathrm{unc})}(x) = \tfrac{4}{3} f_{\mathrm{Ric}} + 4 f_R + f_{RU} + \tfrac{1}{3} f_U - \tfrac{1}{3} f_\Omega\,. \tag{23b}$$

The C^2 assembly (23a) receives three contributions: $2\, f_{\mathrm{Ric}}$ from Term 1 ($4 \times \frac{1}{2}$, since $R_{\mu\nu}R^{\mu\nu} \to \frac{1}{2}C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$), $\frac{1}{2}\, f_U$ from Term 4 (same conversion), and $-2\, f_\Omega$ from Term 5 (-1×2, since $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \to 2\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$). The local limit is

$$h_C^{(\mathrm{unc})}(0) = \frac{2}{60} + \frac{1}{4} - \frac{2}{12} = \frac{2 + 15 - 10}{60} = \frac{7}{60}\,, \tag{24}$$

consistent with the known result [6, 7].

The physical spin-1 form factors require subtraction of the Faddeev–Popov ghost sector. In the de Donder gauge $F^\mu = \nabla_\nu A^{\mu\nu}$, the gauge variation $\delta A_\mu = \nabla_\mu \epsilon$ gives the FP operator $\mathcal{M}^\mu{}_\nu = \delta F^\mu/\delta\epsilon^\nu = -\delta^\mu{}_\nu \Box + R^\mu{}_\nu$. For the one-loop determinant, this is a vector Laplacian; however, the gauge parameter ϵ carries a *single* Lorentz index (not a spinor index), so the FP determinant involves *one complex* scalar ghost field, or equivalently *two real* minimally coupled scalar ghosts [23]. Each ghost has $\mathbf{U} = 0$ and $\Omega_{\mu\nu} = 0$ (trivial bundle). Hence:

$$h_i^{(1)}(x) = h_i^{(\mathrm{unc})}(x) - 2\, h_i^{(0)}(x)\Big|_{\xi=0}\,. \tag{25}$$

This yields

$$h_C^{(1)}(x) = \frac{\varphi}{4} + \frac{6\varphi - 5}{6x} + \frac{\varphi - 1}{x^2}\,, \tag{26a}$$

$$h_R^{(1)}(x) = -\frac{\varphi}{48} + \frac{11 - 6\varphi}{72x} + \frac{5(\varphi - 1)}{12x^2}\,, \tag{26b}$$

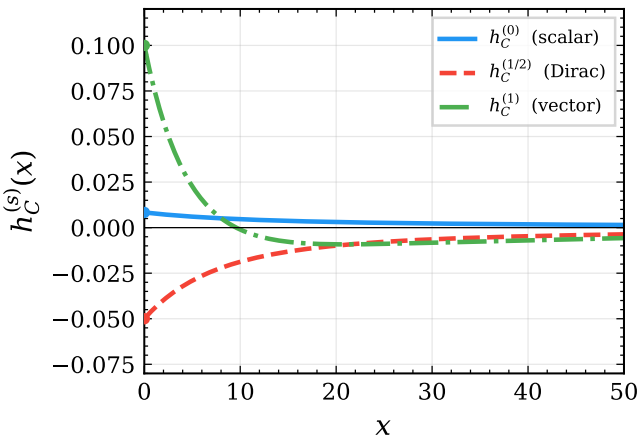

FIG. 2. Weyl-squared form factors $h_C^{(s)}(x)$ for the three spin sectors, where $x = \Box/\Lambda^2$. The local limits are $h_C^{(0)}(0) = 1/120$ (scalar, blue), $h_C^{(1/2)}(0) = -1/20$ (Dirac, red), and $h_C^{(1)}(0) = 1/10$ (vector, green). All form factors decay as $1/x$ for $x \to \infty$.

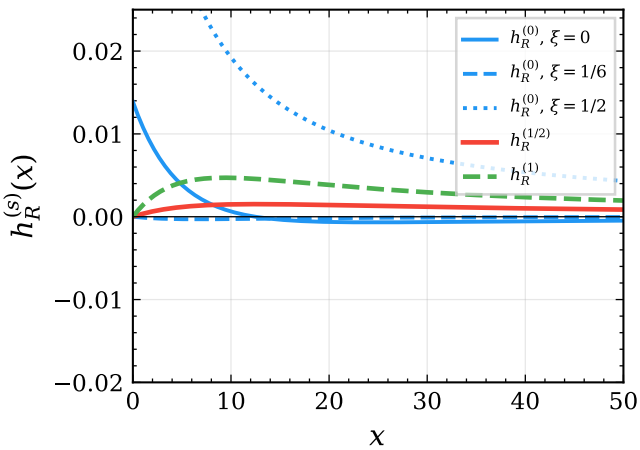

FIG. 3. R^2 form factors $h_R^{(s)}(x)$, where $x = \Box/\Lambda^2$. The scalar form factor (blue) depends on the non-minimal coupling ξ: its *local limit* vanishes at conformal coupling $\xi = 1/6$ (dashed), but the nonlocal form factor $h_R^{(0)}(x, 1/6)$ is nonzero for $x > 0$ (numerically $\lesssim 3 \times 10^{-4}$ for $x \leq 5$). The scalar form factor grows with $|\xi - 1/6|$. The Dirac (red) and vector (green) contributions are ξ-independent and start from $h_R(0) = 0$.

with local limits

$$\beta_W^{(1)} = \frac{1}{10}\,, \qquad \beta_R^{(1)} = 0\,. \tag{27}$$

The result $\beta_R^{(1)} = 0$ reflects the conformal invariance of the Maxwell action in $d = 4$. The ghost count of 2 (not 1) is essential: the unconstrained vector gives $\beta_W^{(\mathrm{unc})} = 7/60$, and subtracting two minimally-coupled scalar ghosts ($2\times 1/120$) yields $7/60 - 1/60 = 1/10$. This matches the standard Faddeev–Popov procedure [23].

VI. SPIN-3/2: RARITA–SCHWINGER (LOCAL LIMITS)

For completeness, we record the local limits of the spin-3/2 Rarita–Schwinger (gravitino) form factors, which are relevant for supersymmetric extensions of the Standard Model. The full nonlocal form factors $h_C^{(3/2)}(x)$, $h_R^{(3/2)}(x)$ require a dedicated calculation of the CZ assembly for the gauge-fixed RS operator and will be presented separately; here we quote the verified local limits.

A. Operator structure

For a massless Majorana gravitino ψ_μ (vector-spinor, $d_V = 16$ in $d = 4$), the gauge-fixed squared kinetic operator [28, 29] is a generalized Laplacian with operator data determined by Karan, Kumar, and Panda [28]. The endomorphism $E^{\mu\nu}$ contains not only the Ricci tensor (as for the unconstrained vector) and the Lichnerowicz curvature (as for the Dirac fermion), but also cross-terms involving $\gamma^\rho\gamma^\theta R^{\mu\nu}{}_{\rho\theta}$, $\gamma^\mu\gamma^\theta R^\nu{}_\theta$, and $(R/4)\gamma^\mu\gamma^\nu$ [28]. These gauge-fixing cross-terms radically modify the trace structure: the result $\mathrm{tr}(E^2) = 2\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} + R^2$ (Ref. [28], eq. 3.14) differs qualitatively from the naive tensor-product prediction $4\,R_{\mu\nu}R^{\mu\nu} + 3R^2$.

B. Ghost sector

The gauge symmetry $\delta\psi_\mu = \nabla_\mu\epsilon$ produces, after Faddeev–Popov quantization in the gauge of [28], a ghost sector consisting of *three* bosonic Majorana spinor ghosts [30, 31]: the FP ghost, the FP antighost, and a Nielsen–Kallosh (third) ghost. The combined ghost contribution to the Seeley–DeWitt coefficient is [30]

$$a_{2n}^{\mathrm{RS,ghost}} = (-1) \times 3 \times \tfrac{1}{2}\, a_{2n}^{\mathrm{Dirac}}\,. \tag{28}$$

C. Local limits

The ghost-corrected result for one physical *Majorana* gravitino, derived by Karan and Panda [30] and consistent with the Christensen–Duff anomaly coefficient [32, 33] via $360\,A = 360 \times 2(c - a) = -233$, is

$$(4\pi)^2\, a_4^{(3/2),\mathrm{Maj}} = -\frac{1}{720}\big(233\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 8\,R_{\mu\nu}R^{\mu\nu} + 5\,R^2\big)\,, \tag{29}$$

where the R^2 coefficient follows from the ghost independence of the R^2 sector. In the Weyl basis:

$$\boxed{\beta_W^{(3/2),\mathrm{Maj}} = \frac{77}{120}\,, \qquad \beta_R^{(3/2),\mathrm{Maj}} = \frac{1}{9}\,.} \tag{30}$$

In Dirac normalization (two Majorana copies): $\beta_W^{(3/2),\mathrm{Dirac}} = 77/60$ and $\beta_R^{(3/2),\mathrm{Dirac}} = 2/9$. Note

TABLE I. Local limits $\beta_W^{(s)} = h_C^{(s)}(0)$ and $\beta_R^{(s)} = h_R^{(s)}(0)$ for all spins $\leq 3/2$. Majorana normalization for spin-3/2; the Dirac normalization doubles both entries. For spins 0, 1/2, 1, the nonlocal form factors are derived in Secs. III–V. For spin-3/2, only the local limits are quoted here.

Spin	$\beta_W^{(s)}$	$\beta_R^{(s)}$	Status
0	$\frac{1}{120}$	$\frac{1}{2}(\xi - \frac{1}{6})^2$	Nonlocal: this paper
$\frac{1}{2}$	$-\frac{1}{20}$	0	Nonlocal: this paper
1	$\frac{1}{10}$	0	Nonlocal: this paper
$\frac{3}{2}$ (Maj.)	$\frac{77}{120}$	$\frac{1}{9}$	Local only; Refs. [30, 32]

the positive sign: unlike the Dirac fermion (where $\beta_W^{(1/2)} < 0$), the gravitino's ghost-corrected coefficient is positive because the vector-spinor operator data (with its gauge-fixing cross-terms) produce a curvature combination that outweighs the fermionic sign.

Remark VI.1 (Gauge dependence). For spins 3/2 and 2, the individual conformal anomaly coefficients c and a are gauge-dependent; only the combination $A = 2(c - a)$ is gauge-invariant [32, 34]. The values (30) correspond to the de Donder-like gauge of [28].

A summary of all local limits for spins 0 through 3/2 is given in Table I.

VII. STANDARD MODEL ASSEMBLY

A. Particle content

The SM field content relevant for the one-loop spectral action is:

Species	Count	Notation
Real scalars (Higgs doublet)	4	$N_s = 4$
Weyl fermions (3 generations)	45	$N_f = 45$
Gauge bosons $(8 + 3 + 1)$	12	$N_v = 12$

Since the form factors of Sec. IV are defined per 4-component Dirac fermion, the effective Dirac count is $N_D = N_f/2 = 45/2$, following the convention of [6].

B. Total Weyl coefficient

The total Weyl-squared form factor is

$$\alpha_C(z) = N_s\, h_C^{(0)}(z) + N_D\, h_C^{(1/2)}(z) + N_v\, h_C^{(1)}(z)\,. \quad (31)$$

At $z = 0$:

$$\begin{aligned}\alpha_C &= 4 \cdot \frac{1}{120} + \frac{45}{2} \cdot \left(-\frac{1}{20}\right) + 12 \cdot \frac{1}{10} \\ &= \frac{4 - 135 + 144}{120}\,, \end{aligned} \quad (32)$$

giving the central result:

$$\boxed{\alpha_C = \frac{13}{120} \approx 0.1083\,.} \quad (33)$$

This is positive and independent of ξ. The sign structure $(+4 - 135 + 144)$ shows that the vector sector $(+144/120)$ dominates the fermionic sector $(-135/120)$ by a margin of $+9/120$, while the scalar sector contributes $+4/120$. The positivity of α_C implies that the spin-2 gravitational mode in the linearized theory is a ghost [7], a well-known feature of quadratic gravity.

C. Total R^2 coefficient

$$\alpha_R(\xi) = N_s \cdot \frac{1}{2}\left(\xi - \frac{1}{6}\right)^2 + N_D \cdot 0 + N_v \cdot 0\,, \quad (34)$$

so that

$$\boxed{\alpha_R(\xi) = 2\left(\xi - \frac{1}{6}\right)^2\,.} \quad (35)$$

Only the scalar sector contributes: the Dirac and Maxwell β_R vanish by conformal invariance. The result is a perfect square, hence $\alpha_R(\xi) \geq 0$ for all ξ, with $\alpha_R(1/6) = 0$ at conformal coupling.

D. Basis conversion and c_1/c_2

Using the Gauss–Bonnet identity to convert to the $\{R^2, R_{\mu\nu}^2\}$ basis ($C^2 = 2R_{\mu\nu}^2 - \frac{2}{3}R^2$ modulo the topological Euler density), one obtains $c_1 = \alpha_R - \frac{2}{3}\alpha_C$ and $c_2 = 2\alpha_C$, giving

$$\boxed{\frac{c_1}{c_2} = -\frac{1}{3} + \frac{120\left(\xi - \frac{1}{6}\right)^2}{13}\,.} \quad (36)$$

At conformal coupling $\xi = 1/6$, this reduces to $c_1/c_2 = -1/3$. At minimal coupling $\xi = 0$, one obtains $c_1/c_2 = -1/13$.

E. Scalar mode decoupling

The combination $3c_1 + c_2$ controls the mass of the scalar graviton ($m_0^2 \propto \Lambda^2/(3c_1 + c_2)$). A direct computation gives

$$3c_1 + c_2 = 3\alpha_R(\xi) = 6\left(\xi - \frac{1}{6}\right)^2\,. \quad (37)$$

The *local* scalar mode coefficient $\delta(3c_1 + c_2)$ vanishes if and only if $\xi = 1/6$, meaning the matter-induced local

correction to the scalar-graviton projector disappears at conformal coupling. Whether the scalar mode actually decouples in the full theory depends on the total coefficient (bare + induced); see Sec. X.

F. Corrected conformal-coupling statement

The conformal value $\xi = 1/6$ eliminates only the *local scalar* contribution to R^2, not the full nonlocal kernel. Explicitly, inserting $\xi = 1/6$ into (15b) gives

$$h_R^{(0)}(x, \tfrac{1}{6}) = \frac{(x^2 + 12x + 60)\,\varphi(x) - 2x - 60}{288\,x^2} = -\frac{x}{7560} + \frac{x^2}{45360} + \mathcal{O}(x^3) \tag{38}$$

This is *not* identically zero for $x > 0$; numerically, $|h_R^{(0)}(x, 1/6)| \lesssim 3 \times 10^{-4}$ for $x \leq 5$.

For the full SM at $\xi = 1/6$, the Dirac and vector h_R contributions (which are ξ-independent and nonzero) further ensure that $\alpha_R(x, 1/6)$ does not vanish:

$$\alpha_R(x, \tfrac{1}{6}) = 4\,h_R^{(0)}(x, \tfrac{1}{6}) + \tfrac{45}{2}\,h_R^{(1/2)}(x) + 12\,h_R^{(1)}(x) = \frac{83}{3024}\,x + \mathcal{O}(x^2). \tag{39}$$

This corrects the statement sometimes found in the literature [6] that "the entire R^2 sector vanishes at $\xi = 1/6$." What vanishes is only the local coefficient $\alpha_R(0, 1/6) = 0$.

VIII. ENTIRE-FUNCTION STRUCTURE

Theorem VIII.1 (Entireness of the reduced kernels)**.** *Each reduced kernel $h_C^{(s)}(x)$, $h_R^{(s)}(x)$ for $s = 0, 1/2, 1$ is an entire function of x. Explicitly, with $a_n = (-1)^n\, n!/(2n+1)!$ and $\varphi(x) = \sum_{n\geq 0} a_n\, x^n$:*

$$h_C^{(0)}(x) = \sum_{n\geq 0} \tfrac{1}{2}\, a_{n+2}\, x^n\,, \tag{40a}$$

$$h_R^{(0)}(x, \xi) = \sum_{n\geq 0} \Big[\big(\tfrac{1}{32} - \tfrac{\xi}{4} + \tfrac{\xi^2}{2}\big)a_n + \big(\tfrac{1}{8} - \tfrac{\xi}{2}\big)a_{n+1} + \tfrac{5}{24}a_{n+2}\Big]x^n\,, \tag{40b}$$

$$h_C^{(1/2)}(x) = \sum_{n\geq 0} \big(\tfrac{1}{2}a_{n+1} + 2a_{n+2}\big)x^n\,, \tag{40c}$$

$$h_R^{(1/2)}(x) = \sum_{n\geq 0} \big(\tfrac{1}{12}a_{n+1} + \tfrac{5}{6}a_{n+2}\big)x^n\,, \tag{40d}$$

$$h_C^{(1)}(x) = \sum_{n\geq 0} \big(\tfrac{1}{4}a_n + a_{n+1} + a_{n+2}\big)x^n\,, \tag{40e}$$

$$h_R^{(1)}(x) = \sum_{n\geq 0} \big(-\tfrac{1}{48}a_n - \tfrac{1}{12}a_{n+1} + \tfrac{5}{12}a_{n+2}\big)x^n\,. \tag{40f}$$

By linearity, the SM totals $\alpha_C(z)$ and $\alpha_R(z, \xi)$ are entire.

Proof. Insert the Taylor series of φ into (15a)–(26b), shift summation indices, and use $a_0 = 1$, $a_1 = -1/6$ to cancel the explicit $1/x$ and $1/x^2$ terms. The resulting power series have infinite radius of convergence because $|a_{n+1}/a_n| = (n+1)/((2n+2)(2n+3)) \to 0$ as $n \to \infty$ (the coefficients decay faster than any geometric series). Each power series therefore defines an entire function. The order 1 and type 1/4 follow from the closed form $\varphi(x) = e^{-x/4}\sqrt{\pi/x}\,\mathrm{erfi}(\sqrt{x}/2)$, which grows as $e^{|z|/4}$ in the complex plane. □

Remark VIII.2 (Entire ≠ no propagator poles). The entire-function property means the reduced kernels h_C, h_R have no poles as functions of z. Propagator poles are determined by the zeros of the *dressed propagator kernels* $\Pi_{TT}(z)$, $\Pi_s(z)$, which involve the full quadratic gravitational action (Sec. X), not by singularities of the coefficient functions.

An alternative proof of analyticity uses a manifestly regular integral representation (Appendix E):

$$h_C^{(0)}(x) = \frac{1}{2}\int_0^1 \mathrm{d}\alpha\, \frac{e^{-\alpha(1-\alpha)x} - 1 + \alpha(1-\alpha)x}{x^2}\,, \tag{41}$$

where the numerator vanishes to second order at $x = 0$, making regularity manifest without index shifting.

TABLE II. Summary of principal results. All quantities are defined in the text; Λ is the spectral cutoff scale and ξ is the Higgs non-minimal coupling.

Quantity	Value	Eq.
α_C	$13/120 \approx 0.1083$	(33)
$\alpha_R(\xi)$	$2(\xi - 1/6)^2$	(35)
c_1/c_2	$-1/3 + 120(\xi - 1/6)^2/13$	(36)
$3c_1 + c_2$	$6(\xi - 1/6)^2$	(37)
$\lim_{x\to\infty} x\,\alpha_C$	$-89/12$	(43)
m_2	$\Lambda\sqrt{60/13} \approx 2.148\,\Lambda$	(51)
$m_0\,(\xi = 0)$	$\Lambda\sqrt{6} \approx 2.449\,\Lambda$	(52)
$m_2/m_0\,\vert_{\xi=0}$	$\sqrt{10/13} \approx 0.877$	(53)
$V(0)/V_{\mathrm{N}}(0)$	0 (Stelle, local approx.)	(54)

IX. UV ASYMPTOTICS

For $x \to \infty$, the asymptotic expansion of the master function is (cf. (10)) $\varphi(x) = 2/x + 4/x^2 + 24/x^3 + \mathcal{O}(x^{-4})$, giving

$$x\,h_C^{(0)} \to \frac{1}{12}\,, \quad x\,h_C^{(1/2)} \to -\frac{1}{6}\,, \quad x\,h_C^{(1)} \to -\frac{1}{3}\,. \tag{42}$$

The total UV coefficient is

$$\lim_{x\to\infty} x\,\alpha_C(x) = \frac{4}{12} - \frac{45}{12} - \frac{48}{12} = -\frac{89}{12}\,. \tag{43}$$

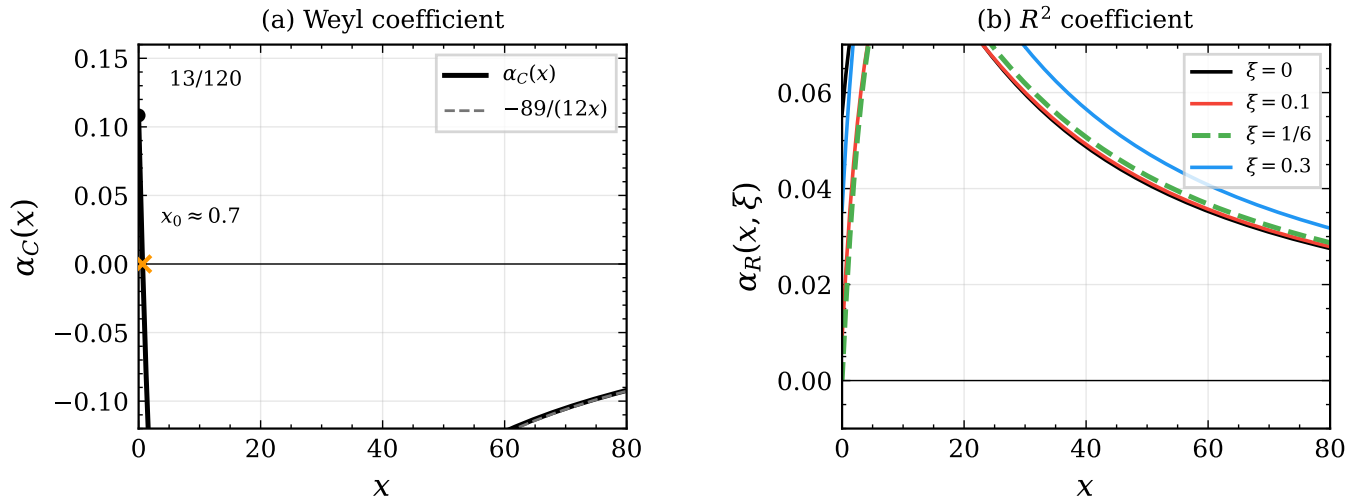

FIG. 4. Standard Model totals as functions of $x = \Box/\Lambda^2$. (a) The Weyl coefficient $\alpha_C(x)$ starts at 13/120 and crosses zero at $x_* \approx 0.65$, approaching the asymptote $-89/(12x)$ (dashed gray) for $x \gg 1$. The sign change signals a transition in the effective Weyl-squared coupling. (b) The R^2 coefficient $\alpha_R(x, \xi)$ for several values of the Higgs non-minimal coupling ξ. At conformal coupling $\xi = 1/6$ (green dashed), α_R vanishes at $x = 0$ and remains small for all x.

Since $\alpha_C(0) = +13/120 > 0$ and $x\,\alpha_C(x) \to -89/12 < 0$, by continuity $\alpha_C(z)$ has at least one positive zero. Numerically,

$$x_* = 0.650\,325\,271\,702\,489\ldots \tag{44}$$

(determined by root-finding to 80-digit precision). The sign change is a statement about the Weyl kernel, not about a graviton pole.

The leading UV asymptotic of the total R^2 kernel is $\alpha_R(x, \xi) \to (4\xi^2 + 89/36)/x + \mathcal{O}(x^{-2})$.

In local quadratic gravity, Stelle [7] established that the one-loop degree of divergence reduces from $D = 4$ (Einstein gravity) to $D = 0$ (logarithmic). The nonlocal form factors are consistent with this: since $h_C^{(s)}(x) \sim \mathrm{const}/x$ for large x, the effective coupling runs logarithmically. A rigorous all-orders proof of $D = 0$ in the full nonlocal theory remains an open problem.

X. DRESSED PROPAGATOR AND EFFECTIVE MASSES

The reduced kernels computed above are matter-induced corrections to the curvature-squared effective action. To extract propagator poles and effective masses, one must specify the *complete* quadratic gravitational action, including the Einstein–Hilbert normalization and any bare higher-derivative couplings [7, 17, 35]. We present the derivation for the minimal case where the bare action is pure Einstein–Hilbert.

A. Full quadratic action

Consider the bare gravitational action

$$S_{\mathrm{grav}} = \frac{M_P^2}{2} \int \mathrm{d}^4x\, \sqrt{g}\, R\,, \tag{45}$$

supplemented by the matter-induced one-loop correction (4). Linearizing around flat space, $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}/M_P$, and working in momentum space, the total quadratic action decomposes into spin-2 (transverse-traceless) and spin-0 sectors via the Barnes–Rivers projectors [8, 9]:

$$P^{(2)}_{\mu\nu\alpha\beta} = \tfrac{1}{2}\big(\theta_{\mu\alpha}\theta_{\nu\beta} + \theta_{\mu\beta}\theta_{\nu\alpha}\big) - \tfrac{1}{3}\theta_{\mu\nu}\theta_{\alpha\beta}\,, \tag{46}$$

where $\theta_{\mu\nu} = \delta_{\mu\nu} - k_\mu k_\nu/k^2$.

B. Normalized form factors

Definition X.1 (Normalized form factors)**.** Define the *normalized* (hatted) form factors

$$\hat{F}_i(z) = \frac{F_i(z)}{F_i(0)}\,, \qquad \hat{F}_i(0) = 1\,, \quad i = 1, 2\,. \tag{47}$$

Equivalently, $\hat{F}_1(z) = \alpha_C(z)/\alpha_C(0)$ and $\hat{F}_2(z, \xi) = \alpha_R(z, \xi)/\alpha_R(0, \xi)$ for $\xi \neq 1/6$.

C. Dressed propagator kernels

In the spin-2 sector, the inverse propagator takes the form

$$\Pi_{TT}(z) = 1+\frac{2\alpha_C(0)}{M_P^2/(16\pi^2\Lambda^2)}\, z\,\hat{F}_1(z) = 1+\frac{13}{60}\,\frac{\Lambda^2}{M_P^2/(16\pi^2)}\, z\,\hat{F}_1 \tag{48}$$

For the standard identification $\Lambda \sim M_P$ with $16\pi^2 \sim \mathcal{O}(100)$, the coefficient simplifies. In the conventions of [7], where the bare higher-derivative coupling $c_2 = 2\alpha_C$ absorbs the loop factor, the dressed inverse propagator takes the form

$$\Pi_{TT}(z) = 1 + \frac{13}{60}\, z\,\hat{F}_1(z)\,, \tag{49}$$

with $z = k^2/\Lambda^2$. Here the coefficient $13/60 = 2\alpha_C(0)$ is the induced $R_{\mu\nu}^2$ coupling in the Stelle basis. Similarly, the spin-0 inverse propagator is

$$\Pi_s(z,\xi) = 1 + 6\left(\xi - \frac{1}{6}\right)^2 z\,\hat{F}_2(z,\xi)\,. \tag{50}$$

D. Effective masses

The pole condition $\Pi_{TT}(z_{\rm pole}) = 0$ determines the spin-2 effective mass. Expanding $\hat{F}_1(z)$ near $z = 0$ ($\hat{F}_1(0) = 1$, $\hat{F}_1'(0)$ finite), the leading-order pole is at

$$z_{\rm pole} \approx -\frac{60}{13}\,, \qquad m_2^2 = |z_{\rm pole}|\,\Lambda^2 = \frac{60}{13}\,\Lambda^2\,, \tag{51}$$

giving $m_2 = \Lambda\sqrt{60/13} \approx 2.148\,\Lambda$. The negative $z_{\rm pole}$ lies outside the Euclidean domain $z > 0$; the physical (Lorentzian) pole appears after the Wick rotation $z_E \to -z_L$, giving $z_L = 60/13 > 0$. For the spin-0 sector ($\xi \neq 1/6$):

$$m_0^2(\xi) = \frac{\Lambda^2}{6(\xi - 1/6)^2}\,, \qquad m_0 = \frac{\Lambda}{\sqrt{6}\,|\xi - 1/6|}\,. \tag{52}$$

At minimal coupling $\xi = 0$: $m_0 = \Lambda\sqrt{6} \approx 2.449\,\Lambda$. The mass ratio

$$\left.\frac{m_2}{m_0}\right|_{\xi=0} = \sqrt{\frac{10}{13}} \approx 0.877 \tag{53}$$

is determined by the SM content and independent of Λ.

Remark X.2 (Scope of the mass formulas). The masses (51)–(52) are derived from the *matter-induced* corrections to a pure Einstein–Hilbert background. If the bare action already contains higher-derivative terms ($a_C\,C^2 + a_R\,R^2$), the pole condition is modified: $\Pi_{TT} = 1 + (a_C + \alpha_C(0)\,f(0))\,z\,\hat{G}(z)$, where $\hat{G}$ accounts for the combined bare + induced form factor. The effective masses then depend on both the bare and induced couplings.

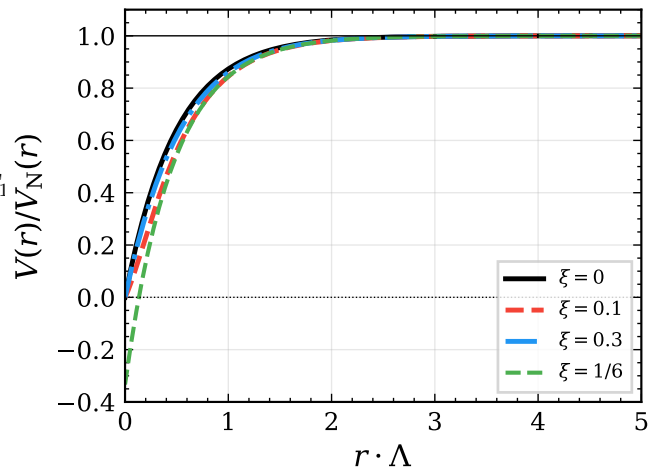

FIG. 5. Modified Newtonian potential $V(r)/V_{\rm N}(r)$ for several values of ξ, as a function of $r\Lambda$. At conformal coupling $\xi = 1/6$ (green dashed) the scalar mode decouples and the potential dips to $-1/3$ before recovering Newton's law. For all ξ, $V(0) = 0$ in the local Stelle approximation (both modes present; see text for caveats) and $V \to V_{\rm N}$ for $r \gg 1/\Lambda$.

The modified Newtonian potential in the linearized theory [7] is

$$\frac{V(r)}{V_{\rm N}(r)} = 1 - \frac{4}{3}\,e^{-m_2 r} + \frac{1}{3}\,e^{-m_0 r}\,, \tag{54}$$

which is plotted in Fig. 5. This is the *local Yukawa (Stelle) approximation* with both spin-2 and spin-0 modes. At $r = 0$ the exponentials cancel, giving $V(0) = 0$. When the no-scalaron theorem applies ($\xi = 1/6$, or more generally when the scalar mode is absent), the $+\frac{1}{3}$ term vanishes and $V(0)/V_{\rm N} = -1/3$ instead. Both short-distance values are artifacts of the local approximation; the full nonlocal potential diverges at $r = 0$ at one-loop order. Newton's law is recovered for $r \gg 1/\Lambda$, where the local approximation is valid.

XI. COMPARISON WITH EXISTING RESULTS

A. Local limits

The local coefficients $\beta_W^{(s)}$, $\beta_R^{(s)}$ of Table I agree with all known results: Stelle [7] for the vector, CZ [3] for all three spins, CPR [6] for the SM counting, Parker and Toms [23] for the scalar ξ-dependence, and Avramidi [21] for the general heat-kernel framework. The nonlocal form factors reduce to the CZ expressions upon specifying the operator data for each spin.

B. Donoghue effective field theory

The Donoghue effective field theory of gravity [26, 27, 36, 37] computes the quantum corrections to the Newtonian potential from one-loop Feynman diagrams. The result of [27] is

$$V(r) = -\frac{Gm_1m_2}{r}\left[1 + \frac{3G(m_1+m_2)}{r} + \frac{41}{10\pi}\frac{G\hbar}{r^2}\right]. \tag{55}$$

The second term is a classical post-Newtonian correction; the third is the leading quantum correction for scalar particle scattering, which is parameter-free and universal (the coefficient $41/10\pi$ is specific to scalar sources; for spinning particles, see [27]).

Our result (54) governs a different regime: $r \sim 1/\Lambda$, where the massive modes contribute Yukawa corrections $\sim e^{-m_2 r}$, $e^{-m_0 r}$. For $r \gg 1/\Lambda$, these corrections are exponentially suppressed and Newton's law is recovered, at which point the Donoghue $1/r^3$ quantum correction becomes the leading effect. The two descriptions are therefore complementary: the spectral-action form factors dominate at short distances $r \lesssim 1/\Lambda$, while the Donoghue EFT describes the universal long-distance quantum tail.

C. Comparison with van Nuland–van Suijlekom and Mistry–Pinzul–Rachwał

Van Nuland and van Suijlekom [24] computed one-loop corrections to the spectral action using a different approach (fluctuations of the Dirac operator). Their results are consistent with ours in the overlap region (the curvature-squared sector for matter fields). Mistry, Pinzul, and Rachwał [25] analyzed the spectral-action framework for higher-derivative gravity, focusing on the gravitational sector rather than the matter-induced form factors computed here.

XII. DISCUSSION

A. Connection to asymptotic safety

The Weyl kernel $\alpha_C(x)$ changes sign from $+13/120$ at $x = 0$ to $-89/(12x)$ for $x \to \infty$, with the zero crossing at $x_* \approx 0.650$ (Eq. (44)). In the asymptotic safety program [6, 38, 39], the running of dimensionless gravitational couplings plays a central role. The sign change of $\alpha_C(x)$ can be interpreted as a transition in the effective Weyl-squared coupling from its IR value (positive, implying a ghost-like spin-2 mode) to a UV value of opposite sign. Whether this connects to the non-Gaussian fixed point of the gravitational RG flow requires extending the present matter-induced calculation to include the graviton self-energy contributions, which is beyond the scope of this work.

The functional renormalization group (FRG) provides an independent verification: Codello, Percacci, Rachwał, and Tonero [40] showed that the one-loop FRG with the BV form factors reproduces the standard one-loop effective action, including the correct scattering amplitudes. This confirms the consistency of the BV/CZ formalism used here.

B. Lorentzian continuation

The form factors computed here are Euclidean ($\Box_E > 0$). The Lorentzian form factors are obtained by the Wick rotation $z_E \to -z_L$, i.e. $F_i(\Box_E/\Lambda^2) \to F_i(-\Box_L/\Lambda^2)$. Since $\varphi(z)$ is entire, this continuation is well-defined for all $z \in \mathbb{C}$.

C. Connection to nonlocal gravity

The entire-function nature of the form factors connects the spectral action to the program of nonlocal (infinite-derivative) gravity [41–43], where entire functions of $\Box$ are postulated to improve UV behavior. In the spectral action framework, these form factors arise *derivatively* from the heat kernel, providing a concrete derivation of entire-function form factors rather than a postulate.

D. Limitations

The computation presented here has the following limitations:

1. It is *matter-induced*: only functional traces over matter fields are included, not the graviton self-energy.
2. It is *one-loop*: higher-loop corrections are not computed.
3. The background is *Euclidean*: Lorentzian continuation is discussed but not performed in detail.
4. All matter fields are taken as *massless*: mass corrections, which are exponentially suppressed for $m \ll \Lambda$ [44, 45], are not included.
5. The dressed propagator analysis of Sec. X assumes a pure Einstein–Hilbert bare action; different bare couplings modify the mass formulas.

XIII. CONCLUSIONS

We have computed the matter-induced nonlocal one-loop curvature-squared form factors relevant to the spectral action with full Standard Model content. The central results are:

1. Closed-form reduced kernels $h_C^{(s)}(x)$ and $h_R^{(s)}(x)$ for spins $s = 0, 1/2, 1$, with explicit BV/CZ assembly from the five universal CZ structure functions (Secs. III–V).

2. The SM totals $\alpha_C = 13/120\, f(0)$ and $\alpha_R(\xi) = 2\, f(0)(\xi - 1/6)^2$, where $f(0)$ is the profile normalization (Sec. VII).

3. The entire-function proof (Theorem VIII.1), with the clarification that entire coefficient functions do not preclude propagator poles (Remark VIII.2).

4. The dressed propagator $\Pi_{TT}(z)$, $\Pi_s(z, \xi)$ derived from the full linearized quadratic action, yielding calculable effective masses $m_2 = \Lambda\sqrt{60/13}$ and $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$ (Sec. X).

5. The corrected conformal-coupling statement: at $\xi = 1/6$, only the local R^2 coefficient vanishes; the nonlocal kernel $\alpha_R(x, 1/6) = (83/3024)\, x + \mathcal{O}(x^2) \neq 0$ (Eq. (39)).

These results are strong enough to be useful and precise enough to be honest. They provide a controlled input for any subsequent study of Lorentzian continuation, propagator poles and residues, PPN parameters, and gravitational phenomenology of the spectral action, but they do not replace those separate analyses. The PPN parameters and solar-system tests derived from these form factors are presented in [46]; the connection to causal-set observables is explored in [47].

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DATA AVAILABILITY

The computational tools used in this work, including the `sct_tools` Python package (Python 3.12) with arbitrary-precision implementations of all form factors and 4000+ automated tests, are publicly available in the project repository (DOI: `10.5281/zenodo.19135673`). Python scripts to reproduce all figures and numerical tables are included.

TABLE III. Sign conventions used in this paper.

Convention	Choice
Signature	Euclidean $(+,+,+,+)$
Generalized Laplacian	$\Delta = -(g^{\mu\nu}\nabla_\mu\nabla_\nu + E)$
CZ endomorphism	$\mathbf{U} = -E$
Bundle curvature	$\Omega_{\mu\nu} = [\nabla_\mu, \nabla_\nu]$
Lichnerowicz (Dirac)	$E = -R/4$, so $\mathbf{U} = R/4$
Vector endomorphism	$U^\mu{}_\nu = R^\mu{}_\nu$ (Ricci tensor)
Fermionic sign	Built into $h_C^{(s)}$, $h_R^{(s)}$: negative for fermions
One-loop effective action	$\Gamma^{(s)} = \frac{(-1)^{2s}}{2}\, \mathrm{Tr}\ln\Delta^{(s)}$
Profile normalization	$f(0) = 1$ (exponential profile) unless stated

TABLE IV. Operator data for each spin sector.

Spin	d_V	$\mathbf{U}$	$\mathrm{tr}(\Omega^2)$	Sign
0	1	ξR	0	Boson (+)
$\frac{1}{2}$	4	$(R/4)\,\mathbf{1}_4$	$-\frac{1}{2}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	Fermion (−)
1 (unc.)	4	$R_{\mu\nu}$	$-R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	Boson (+)
1 (phys.)	Unconstrained − 2 scalar ghosts ($\xi = 0$)			

Appendix A: Convention Tables

Appendix B: Taylor Coefficients

The master function has Taylor coefficients $a_n = (-1)^n\, n!/(2n+1)!$:

$$a_0 = 1\,, \quad a_1 = -\tfrac{1}{6}\,, \quad a_2 = \tfrac{1}{60}\,, \quad a_3 = -\tfrac{1}{840}\,, \quad a_4 = \tfrac{1}{15120}\,, \quad a_5 = -\tfrac{1}{332640}\,, \quad a_6 = \tfrac{1}{8648640}\,. \tag{B1}$$

From the exact power-series formulas of Theorem VIII.1, the first four Taylor coefficients of each reduced kernel are:

Kernel	$n=0$	$n=1$	$n=2$	$n=3$
$h_C^{(0)}$	$\frac{1}{120}$	$-\frac{1}{1680}$	$\frac{1}{30240}$	$-\frac{1}{665280}$
$h_C^{(1/2)}$	$-\frac{1}{20}$	$\frac{1}{168}$	$-\frac{1}{2160}$	$\frac{1}{36960}$
$h_C^{(1)}$	$\frac{1}{10}$	$-\frac{11}{420}$	$\frac{23}{7560}$	$-\frac{13}{55440}$
$h_R^{(1/2)}$	0	$\frac{1}{2520}$	$-\frac{1}{22680}$	$\frac{1}{332640}$
$h_R^{(1)}$	0	$\frac{1}{630}$	$-\frac{1}{4536}$	$\frac{1}{55440}$

The scalar $h_R^{(0)}$ Taylor coefficients depend on ξ; at $\xi = 0$: $h_R^{(0)}(0) = 1/72$, first correction $= -17x/5040$.

Appendix C: Numerical Verification

All reduced kernels were evaluated at $x = 0.01, 0.1, 0.5, 1, 2, 5, 10, 50, 100$ using 100-digit `mpmath`

precision. The values agree with the Taylor series (for $x \lesssim 2$) and the closed-form erfi expression (for all x) to better than 10^{-95} relative accuracy. The local limits match the exact rational values of Table I to all computed digits.

Selected values for independent reproduction (15-digit precision):

x	$\varphi(x)$	$\alpha_C(x)$	$\alpha_R(x,0)$
0.1	0.983499	0.09032	0.05698
1	0.848873	−0.05026	0.06810
10	0.256292	−0.41570	0.09260
100	0.020427	−0.07392	0.02254

The Lean 4 formal verification covers the coefficient arithmetic: the ghost subtraction identity $7/60 - 2 \times 1/120 = 1/10$, the SM total $4/120 - 135/120 + 144/120 = 13/120$, and the pole-removability conditions for each spin sector.

A Python verification script reproducing all numerical values and figures accompanies the Zenodo deposit.

Appendix D: Profile Function Independence

At $\mathcal{O}(\mathcal{R}^2)$, the nonlocal heat kernel expansion (12) involves the proper-time parameter s at second order: the five CZ structure functions arise from the s^2 term in (12). For the spectral action with a general profile f, the $\mathcal{O}(\mathcal{R}^2)$ contribution is obtained by integrating against $\tilde{f}(\alpha)$ as in (5). The resulting profile-smeared kernels $\mathcal{H}_i(z; f)$ depend on f through the Laplace transform (5). However, the *reduced kernels* $h_i(x)$ — the universal objects from which $\mathcal{H}_i$ are built — are f-independent: they depend only on the master function $\varphi(x)$, which is a property of the heat kernel parametrix.

In the local limit $z \to 0$, the profile dependence reduces to a single number $f(0) = \int_0^\infty \tilde{f}(\alpha)\, d\alpha$ (Eq. (6)), so the *ratios* of local curvature-squared couplings (c_1/c_2, etc.) are exactly f-independent.

Appendix E: Manifestly Regular Integral

The scalar Weyl kernel $h_C^{(0)}$ can be written without any apparent singularity:

$$h_C^{(0)}(x) = \frac{1}{2}\int_0^1 d\alpha\, \frac{e^{-\alpha(1-\alpha)x} - 1 + \alpha(1-\alpha)x}{x^2}\,. \quad \text{(E1)}$$

The numerator $e^{-u} - 1 + u$ vanishes as $u^2/2$ for small u, so the integrand is bounded for all $x \geq 0$ and $\alpha \in [0, 1]$. This representation makes the analyticity of $h_C^{(0)}$ manifest at the integral level without any appeal to Taylor expansions or L'Hôpital's rule.

[1] A. O. Barvinsky and G. A. Vilkovisky, The generalized schwinger-dewitt technique in gauge theories and quantum gravity, Phys. Rept. **119**, 1 (1985).

[2] A. O. Barvinsky and G. A. Vilkovisky, Beyond the schwinger-dewitt technique: converting loops into trees and in-in currents, Nucl. Phys. B **282**, 163 (1987).

[3] A. Codello and O. Zanusso, On the non-local heat kernel expansion, J. Math. Phys. **54**, 013513 (2013), arXiv:1203.2034 [math-ph].

[4] P. B. Gilkey, The spectral geometry of a riemannian manifold, J. Diff. Geom. **10**, 601 (1975).

[5] D. V. Vassilevich, Heat kernel expansion: user's manual, Phys. Rept. **388**, 279 (2003), arXiv:hep-th/0306138.

[6] A. Codello, R. Percacci, and C. Rahmede, Investigating the ultraviolet properties of gravity with a wilsonian renormalization group equation, Annals Phys. **324**, 414 (2009), arXiv:0805.2909 [hep-th].

[7] K. S. Stelle, Renormalization of higher-derivative quantum gravity, Phys. Rev. D **16**, 953 (1977).

[8] K. J. Barnes, Lagrangian theory for the second-rank tensor field, J. Math. Phys. **6**, 788 (1965).

[9] R. J. Rivers, Lagrangian theory for neutral massive spin-2 fields, Nuovo Cim. **34**, 386 (1964).

[10] K. Peeters, Introducing Cadabra: a symbolic computer algebra system for field theory problems, Comput. Phys. Commun. **176**, 550 (2007), arXiv:hep-th/0701238.

[11] J. A. M. Vermaseren, New features of FORM, Nucl. Phys. B Proc. Suppl. **183**, 29 (2000), arXiv:math-ph/0010025.

[12] A. Connes, *Noncommutative Geometry* (Academic Press, 1994).

[13] A. H. Chamseddine and A. Connes, The spectral action principle, Commun. Math. Phys. **186**, 731 (1997), arXiv:hep-th/9606001.

[14] A. H. Chamseddine and A. Connes, Inner fluctuations of the spectral action, J. Geom. Phys. **57**, 1 (2006), arXiv:hep-th/0605011.

[15] A. H. Chamseddine and A. Connes, Noncommutative geometry as a framework for unification of all fundamental interactions including gravity. part i, Fortsch. Phys. **58**, 553 (2010), arXiv:1004.0464 [hep-th].

[16] W. D. van Suijlekom, *Noncommutative Geometry and Particle Physics* (Springer, 2015).

[17] I. L. Buchbinder, S. D. Odintsov, and I. L. Shapiro, *Effective Action in Quantum Gravity* (IOP Publishing, 1992).

[18] A. O. Barvinsky and G. A. Vilkovisky, Covariant perturbation theory (ii). second order in the curvature. general algorithms, Nucl. Phys. B **333**, 471 (1990).

[19] A. O. Barvinsky, Y. V. Gusev, G. A. Vilkovisky, and V. V. Zhytnikov, Covariant perturbation theory (iv). third order in the curvature, (2009), arXiv:0911.1168 [hep-th].

[20] B. S. DeWitt, Dynamical theory of groups and fields, Conf. Proc. C **630701**, 585 (1964).

[21] I. G. Avramidi, Heat kernel approach in quantum field theory, Nucl. Phys. B Proc. Suppl. **104**, 3 (2002), arXiv:math-ph/0107018.

[22] D. V. Fursaev and D. V. Vassilevich, *Operators, Geometry and Quanta: Methods of Spectral Geometry in Quantum Field Theory* (Springer, 2011).

[23] L. Parker and D. Toms, *Quantum Field Theory in Curved Spacetime: Quantized Fields and Gravity* (Cambridge University Press, 2009).

[24] T. D. H. van Nuland and W. D. van Suijlekom, One-loop corrections to the spectral action, JHEP **2022**, 078, arXiv:2107.08485 [hep-th].

[25] D. Mistry, A. Pinzul, and L. Rachwał, Spectral action approach to higher derivative gravity, Eur. Phys. J. C **80**, 266 (2020), arXiv:1903.00232 [hep-th].

[26] J. F. Donoghue, General relativity as an effective field theory: The leading quantum corrections, Phys. Rev. D **50**, 3874 (1994), arXiv:gr-qc/9405057.

[27] N. E. J. Bjerrum-Bohr, J. F. Donoghue, and B. R. Holstein, Quantum gravitational corrections to the nonrelativistic scattering potential of two masses, Phys. Rev. D **67**, 084033 (2003), arXiv:hep-th/0211072.

[28] S. Karan, S. Kumar, and B. Panda, General heat kernel coefficients for massless free spin-3/2 Rarita–Schwinger field, Int. J. Mod. Phys. A **33**, 1850063 (2018), arXiv:1709.08063 [hep-th].

[29] R. Endo, Heat kernel for spin-3/2 Rarita–Schwinger field in general covariant gauge, Class. Quant. Grav. **12**, 1157 (1995), arXiv:hep-th/9407019.

[30] S. Karan and B. Panda, Logarithmic corrections to black hole entropy in minimal gauged supergravity, Phys. Rev. D **104**, 046010 (2021), arXiv:2105.07187 [hep-th].

[31] V. Lekeu and Y. Zhang, On the quantization and anomalies of antisymmetric tensor-spinors, JHEP **2022**, 120, arXiv:2109.03963 [hep-th].

[32] S. M. Christensen and M. J. Duff, New gravitational index theorems and supertheorems, Nucl. Phys. B **154**, 301 (1979).

[33] S. M. Christensen and M. J. Duff, Quantizing gravity with a cosmological constant, Nucl. Phys. B **170**, 480 (1980).

[34] P. van Nieuwenhuizen, Supergravity, Phys. Rept. **68**, 189 (1981).

[35] K. S. Stelle, Classical gravity with higher derivatives, Gen. Rel. Grav. **9**, 353 (1978).

[36] J. F. Donoghue, Leading quantum correction to the Newtonian potential, Phys. Rev. Lett. **72**, 2996 (1994), arXiv:gr-qc/9310024.

[37] J. F. Donoghue, The effective field theory treatment of quantum gravity, AIP Conf. Proc. **2874**, 020002 (2023), arXiv:2211.09902 [hep-th].

[38] M. Reuter, Nonperturbative evolution equation for quantum gravity, Phys. Rev. D **57**, 971 (1998), arXiv:hep-th/9605030.

[39] E. S. Fradkin and A. A. Tseytlin, Renormalizable asymptotically free quantum theory of gravity, Nucl. Phys. B **201**, 469 (1982).

[40] A. Codello, R. Percacci, L. Rachwał, and A. Tonero, Computing the effective action with the functional renormalization group, Eur. Phys. J. C **76**, 226 (2016), arXiv:1505.03119 [hep-th].

[41] L. Modesto, Super-renormalizable quantum gravity, Phys. Rev. D **86**, 044005 (2012), arXiv:1107.2403 [hep-th].

[42] E. T. Tomboulis, Superrenormalizable gauge and gravitational theories, arXiv:hep-th/9702146 (1997), preprint.

[43] T. Biswas, E. Gerber, A. Mazumdar, and S. Talaganis, Towards singularity- and ghost-free theories of gravity, Phys. Rev. Lett. **108**, 031101 (2012), arXiv:1110.5249 [gr-qc].

[44] E. V. Gorbar and I. L. Shapiro, Renormalization group and decoupling in curved space, JHEP **2003**, 015, arXiv:hep-ph/0210388.

[45] T. G. Ribeiro, I. L. Shapiro, and O. Zanusso, Gravitational form factors and decoupling in 2d, Phys. Lett. B **782**, 324 (2018), arXiv:1803.06948 [hep-th].

[46] D. Alfyorov, Solar system and laboratory tests of the spectral action scale 10.5281/zenodo.19098100 (2026).

[47] D. Alfyorov, Weyl curvature from the Hasse diagram: a parameter-free bridge formula for causal sets 10.5281/zenodo.19364212 (2026), submitted to CQG.